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Assessment of cellular iron levels (FerroOrange high-content imaging) and expression of ferritin light chain (immunoblotting) in PPMI iPSCs

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Summary

Cellular iron content (a) and protein expression of iron-related factors (b) were analyzed in iPSCs from LRRK2 mutation carriers and healthy participants in the PPMI cohort. (a) High-content imaging of PPMI iPSC lines stained with BioTracker FerroOrange live cell dye (Dojindo) was performed on the Lionheart FX (BioTek) automated microscope on five healthy controls, five G2019S LRRK2 and two R1441G LRRK2 iPSC lines from the PPMI. FerroOrange is a fluorescent live cell dye that is cell permeable, reacts irreversibly with cellular ferrous iron and has no chelating ability[1]*. (b) Standard immunoblotting protocol was used to asses Ferritin light Chain levels.*

Method

iPSCs were maintained in StemFlex™ medium (+StemFlex Supplement). For cellular iron measurement cells were plated on Greiner Bio-One µClear™ 96 well plates, precoated with Geltrex matrix (Thermo), at 15K cells/well. Six wells per line were plated in a column-wise fashion to limit row-based bias from the horizontal scanning by the automated microscope. The following day media was replaced with 100µl StemFlex™ medium (+StemFlex Supplement) per well containing 1 µg/ml Hoechst 33342 and 1µM FerroOrange (dissolved in DMSO) or DMSO control, and incubated for 30 mins at 37C, 5% CO_2. Media was replaced with 100µl StemFlex™ medium and cells were imaged on the Lionheart FX (BioTek) automated microscope. Four non-overlapping fields were imaged per well using a 20x objective across 6 wells (>1600 cells). Hoechst (350/461 nm) staining was used for auto-focusing on each field (DAPI cube) and FerroOrange (543/580 nm) was imaged using the RFP cube of the Lionheart FX microscope. Total RFP intensity of each field was divided per cell number (nuclei) and average FerroOrange staining per cell was plotted. For assessment of protein levels of iron-related proteins standard immunoblotting protocol was followed on cell lysates made in lysis buffer (Cell Signaling 9803) using a commercially available antibody against ferritin light chain (ab69090).

References

1. FerroOrange Dojingo product page: https://www.dojindo.com/products/F374/

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